

Skander LASSOUED

Full stack developer

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Software Engineer with over 3 years of experience designing and implementing scalable, maintainable systems. Skilled in delivering innovative solutions, optimizing performance, and driving project success across full-stack development.

SKILLS

- **Frontend:** Angular, React, NextJs, React query, Zustand, Tailwind, Shadcn, PrimeNg, ReactPrime, Typescript, Zod, React form hook
- **API Development:** REST API, GraphQL (Code first, Apollo Federation), tRPC
- **Backend & Microservices:** NodeJS, ExpressJS, NestJS, NATS
- **Databases:** PostgreSQL, MySQL, MongoDB, Prisma
- **Cloud & Storage:** AWS S3, OVH S3
- **Version Control:** Git, GitLab, GitHub
- **Testing & Automation:** Jest, CI/CD with GitHub Actions
- **Real-Time Systems:** Socket.io, Subscription (GraphQL)

PROFESSIONAL EXPERIENCE

STEPS - Back-end developer - May 2024 - Feb 2025

Tunis, Tunisia

- Developed **secure user authentication & role management** using Auth0 (Okta) and NestJS.
- Implemented a **notification system** with Nodemailer for transactional emails.
- Integrated **AWS S3** for file storage, enabling efficient media upload and retrieval.
- Optimized **database performance** using Redis caching, reducing API response time.
- Built event-driven **microservices** with NestJS & NATS, improving system scalability.
- Automated tasks using **cron jobs**, ensuring smooth system operations.

MS-TECHSOFT - Full stack developer - Sep 2022 - May 2024

Tunis, Tunisia

Full stack developer

Responsibilities:

- Built applications with **React, Angular & NestJS**, leveraging **Apollo Federation (GraphQL Code-First)**.
- Developed a **real-time tracking system** for equipment monitoring using **Socket.io**.
- Optimized **MongoDB queries**, improving data retrieval efficiency by **40%**.

FIXTRONIX (FREELANCE) - Full stack developer - JAN 2024 - Mar 2024

Tunis, Tunisia

- Developed a **custom task & ticket management system** using React & NestJS.
- Designed **role-based access control (RBAC)** for secure feature management.
- Implemented **JWT authentication** (Access & Refresh Tokens) for session security.
- Built **scalable REST APIs**, ensuring seamless integration and high performance.

PROJECTS

BOSE Project – Microservices & Cloud Architecture

- Developed **user authentication & role management** with Auth0 (Okta).
- Built a **microservices-based system** using **NestJS & NATS** for seamless service communication.
- Implemented **AWS S3 storage** and **Redis caching**, reducing database load by **30%**.
- Automated system tasks with **cron jobs** for better operational efficiency.

Technologies: NestJS, Microservices, NATS, PostgreSQL, MongoDB, Redis, AWS S3, Swagger.

EXCO Project – Code Migration from HapiJS to NestJS

- Migrated the entire backend from **HapiJS to NestJS**, modernizing system architecture.
- Developed **new REST APIs** and optimized database queries for **PostgreSQL & MongoDB**.
- Integrated **Nodemailer for email notifications**, improving user communication.

Technologies: NestJS, NATS, PostgreSQL, MongoDB, Redis, NATS, AWS S3, Swagger

Fund Management Project – AI & Data Processing

- Designed **secure authentication** with Auth0 for **user management**.
- Integrated an **AI-based data matching system** using Ollama.
- Implemented **message queues (BullMQ)** to manage AI processing efficiently.

Technologies: NestJS, NATS, Redis, AWS S3, Okta (Auth0), MongoDB, Ollama, BullMQ

Astrallys Project – Ticket Management System

- Developed a **modern UI** using React, Zustand, React Query & PrimeReact.
- Refactored and optimized existing **components & MySQL queries** for faster execution.
- Implemented a **real-time notification system** for ticket updates.
- **Bug Fixes & Continuous Improvement:** Addressed bugs and continuously improved the system through testing and monitoring.

Technologies: React, NestJS, MySQL, Twilio, OVH S3, Brevo

GMAO Project – Industrial Equipment Management System

- Developed a **full-featured maintenance tracking system** for industrial equipment.
- Built a **task & ticket management system** with React & NestJS.
- Used **Zustand for state management** and **React Query for optimized data fetching**.
- Implemented **secure role-based access control (RBAC)** for different user permissions.
- Optimized **MongoDB performance**, reducing query response times.
- Designed a **JWT authentication system** with access & refresh tokens.

Technologies: React, NestJS, Zustand, React Query, React Hook Form, Zod, Tailwind CSS, GraphQL, Socket.io, JWT, MongoDB

POS Gateway Terminal – Real-Time Transactions

- Developed **real-time transaction tracking** for **electronic payment terminals (TPE)**.
- Integrated **WebSockets (Socket.io)** for instant transaction updates.
- Implemented **secure role & permission management** for user access control.
- Ensured secure and real-time financial transactions for businesses.

Technologies: Angular, NestJS, RxJS, Apollo GraphQL, MongoDB, JWT, Socket.io.

Display Project – Real-Time Digital Content Broadcasting

- Built an **Angular & NestJS** application to **broadcast multimedia content**.
- Integrated **real-time updates** for connected screens using **WebSockets**.
- Optimized system performance for **high concurrent screen connections**.

Technologies: Angular, NestJS, GraphQL (Code-first), MongoDB, Socket.io.

EDUCATION

<i>Software engineer</i>	2019-2022
<i>Math-Physics Preparatory School.</i>	2017 - 2019

LANGUAGE

- **French:** B1 (Intermediate)
- **English:** B1 (Intermediate Proficiency)
- **German:** A1 (Basic)
- **Arabic:** Native